MIL-B-1070E
1 August 1986
SUPERSEDING
MIL-B-1070D
24 February 1965

MILITARY SPECIFICATION

BREAD, WHITE, ENRICHED, CANNED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This document covers canned enriched white bread for use by the Armed Forces as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 *Government documents.*

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-43991 - Canned Baked, Canned Steamed, Subsistence Items, Packaging Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014; by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8920

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-900 - Bacterial Standards for Starches, Flour, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of solicitation.

AMERICAN PHARMACEUTICAL ASSOCIATION

The National Formulary

(Application for copies should be addressed to the American Pharmaceutical Association, 2215 Constitution Ave., Washington, DC 20037.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 111 North 19th Street, Suite 210, Arlington, VA 22209.)

CONTAINER CORPORATION OF AMERICA

Color Harmony Manual

(Application for copies should be addressed to Container Corporation of America, 38 S. Dearborn Street, Chicago, IL 60603.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Flour. The flour shall be matured, bleached, enriched, hard wheat flour which will produce a product in compliance with 3.4. Alternatively, unenriched flour may be used providing the equivalent enrichment required in the Standard of Identity for Enriched Flour (21 CFR, Part 137. 165) is added at the time of production of the finished product.

3.2.2 Water. Water used for formulation and washing shall conform to the National Interim Primary Drinking Water Regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.

3.2.3 Shortening. Shortening shall be refined hydrogenated cottonseed or peanut oil or a combination of both and shall have a stability of not less than 100 hours as determined by the active oxygen method (AOM). Shortening used for greasing dough trough, dough pieces, or can interiors shall conform to the above requirements.

3.2.4 D-sorbitol. The D-sorbitol used shall be of crystalline form and shall comply with the Food Chemicals Codex.

3.2.5 Sugar. Granulated or liquid sugar shall be used.

3.2.5.1 Sugar, granulated. Sugar shall be white, refined, granulated cane or beet sugar or a blend of both and shall comply with MIL-STD-900.

3.2.5.2 Sugar, liquid. Liquid sugar shall be an aqueous solution of cane or beet sugar or any combination thereof and shall comply with MIL-STD-900.

3.2.6 Invert syrup. Invert syrup shall be water white and consist of 77 percent solids. The percent invert shall be 43 percent; the ash shall be 0.02 percent or less; and the percent sucrose shall be 34 percent.

3.2.7 Salt. Salt shall be noniodized, white, refined sodium chloride, with or without anticaking agents.

3.2.8 Yeast. Yeast shall be good quality commercial baker's yeast.

3.2.9 Inactive dry yeast. Inactive dry yeast shall be primary grown yeast or properly debittered brewers yeast complying with the requirements of the National Formulary, N.F. quality except for vitamin potency. Sufficient quantities of fortifying agents shall be added to the yeast during processing to yield a finished product having not less than the following vitamin potencies:

Vitamin	Micrograms per gram
Thiamin	600
Riboflavin	600
Niacin	1000

The yeast shall be specially treated for use in yeast-leavened doughs so as to: (1) minimize the softening action on the dough; (2) reduce the free amino radical in the product to minimize the Maillard reaction with reducing sugar in the finished product. The amino nitrogen shall be not more than 0.5 percent of the inactive dry yeast.

3.2.10 Sodium stearoyl-2-lactylate. Sodium stearoyl-2-lactylate shall be powdered and free flowing and shall comply with the Food Chemicals Codex.

Source: https://assist.dla.mil -- Downloaded: 202640-16T12:38Z
Check the source to verify that this is the currentversion before use.

3.2.11 Lactic acid. The lactic acid shall comply with the Food Chemicals Codex.

3.3 Preparation. The bread shall be manufactured by the straight dough or sponge and dough method in accordance with 3.3.1, 3.3.2, and 3.3.3. The pH in the finished product specified in 3.4.1 shall be attained by using a prolonged sponge fermentation time and by adding lactic acid during the remix operation. The lactic acid shall be mixed with the remix water and shall be used sparingly to achieve the specified pH in 3.4.1. Any other method yielding an equivalent product will be permissible. The dough shall be formulated from the following ingredients in the proportions specified:

Ingredient	Parts by weight
Flour	100.00
Water	Approx. 50.00 (variable)
Shortening	20.00
D-Sorbitol	7.00
Sugar	5.00 (dry basis)
Invert syrup	1.80
Salt	1.70
Yeast	1.60
Inactive dry yeast	1.10
Sodium stearoly-2-lactylate	0.15
Lactic acid	0.25 (variable)

3.3.1 Sponge and dough. The sifted flour, yeast, and water shall be uniformally mixed to form the sponge. After the sponge is properly fermented, it shall be returned to the mixer and the remainder of the ingredients in the form of a paste and a lactic acid/water mixture added. A batch of fresh paste shall be made for each sponge by mixing together the specified amounts (see 3.3) of shortening, sugar, inactive dry yeast, salt, sodium stearoly-2-lactylate, invert syrup, and D-sorbitol. Lactic acid and water shall be mixed together and added as required to provide control of pH and to obtain a more uniform finished product. The dough shall be mixed until properly developed.

3.3.2 Filling and proofing. After final mixing (see 3.3.1), the dough shall be divided into uniform pieces. The dough shall be lightly greased and placed in clean, dry cans. Scaling weights shall be sufficient to give the required amount of baked bread in each can (see MIL-C-43991). The lids of the cans shall be clinched on loosely to allow for the escape of moisture and fermentation gases evolved during proofing and baking. In order to determine the proper proof, the lids on the control cans shall not be clinched but shall be placed on the cans so that they can be easily removed for observation. When the dough pieces in the cans are proofed to the correct height, the cans shall be placed in the oven.

Source: https://assist.dla.mil -- Downloaded: 20261O-16T12:38Z
Check the source to verify that this is the current version before use.

3.3.3 Processing. The bread shall be thoroughly baked to a uniform color. The clinched can shall be hermetically sealed immediately upon removal from the oven and at or above an internal temperature of 200°F. so that the actual vacuum in the can conforms to the requirements of MIL-C-43991. Sealing of the cans shall be timed so that the actual vacuum in the can conforms to the requirements of MIL-C-43991. Air, water spray, or other suitable cooling methods, with the exception being immersion in a water bath, may be used to cool the cans. The internal temperature of the bread in the cans shall be reduced to below 100°F within 1 hour after the cans are hermetically sealed.

3.4 Finished product. The bread shall conform to the inside contour of both the bottom and the side of the can, and the unfilled space within the container above the bread surface shall be not more than 5 percent of the can volume. The bread may touch the top lid, provided it does not touch the lid at any point closer than 1/4 inch from the can side. No portion of the product shall be extruded into the double seam of the can. The crust shall have "bloom" color of not lighter than Hue 1-1/2 db dull side, of Color Harmony Manual of Container Corporation of America. The crumb shall be creamy white. When the baked load is cut in half from top to bottom, no dense crumb compression streak shall be evident beneath the crust. The bread shall have no off-odors, off-flavors, or foreign material present.

3.4.1 pH and moisture content. The pH of the bread shall not exceed 4.8 when tested as specified in 4.5.5. The finished bread shall have a total moisture content of not more than 35.0 percent nor less than 32.0 percent when tested as specified in 4.5.5.

3.4.2 Water activity. Bread crumb taken from within 2 inches of the center of the loaf, at least 7 days after baking, shall have a calculated mean water activity not greater than 0.930 and an unbiased standard deviation for that mean not greater than the applicable standard deviation specified below.

Calculated mean 1/	Maximum unbiased standard deviation	Calculated mean 1/	Maximum unbiased standard deviation
0.930	0.0028	0.865	0.0120
0.925	0.0035	0.860	0.0127
0.920	0.0042	0.855	0.0135
0.915	0.0049	0.850	0.0142
0.910	0.0056	0.845	0.0149
0.905	0.0063	0.840	0.0156
0.900	0.0071	0.835	0.0163
0.895	0.0078	0.830	0.0170
0.890	0.0085	0.825	0.0177
0.885	0.0092	0.820	0.0184
0.880	0.0099	0.815	0.0191
0.875	0.0106	0.810	0.0198
0.870	0.0113	0.805	0.0206

1/ For calculated means lying between tabular values, use the higher tabular value.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.

3.4.3 Net weight. The minimum net weight of baked canned bread, for the applicable can size (see 6.1), shall be as specified in MIL-C-43991.

3.4.4 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging, or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirement of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, US Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Product examination. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects.

TABLE I. Product defects 1/

Examine	Defect
Major	
151	Crust color lighter than specified in 3.4.
152	Crumb not creamy white.
153	When baked loaf is cut in half from top to bottom, dense crumb compression streak is evident beneath the crust.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour), or foreign color shall be cause for rejection of the lot.

4.5.4 Container fill examination. The finished product shall be examined for the defects listed in table II. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for minor defects.

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:38Z
Check the source to verify that this is the currentversion before use.

TABLE II. Container fill defects

Category	Defect
Minor	
201	Bread does not conform to the inside contour of the bottom or side of the can or both.
202	Unfilled space within the container above the bread surface is more than 5 percent of the can volume. 1/
203	Bread touches top lid at a point closer than 1/4 inch from the can side.
204	Portion of the product extruded into double seam of the can.

1/ Determine the percent of unfilled space within the container above the bread by the following procedure or one giving comparable results:

Volume above unopened can lid: Fill the space above unopened can lid with rape seed and level off the top of can by use of a straight edge. Measure volume of rape seed.

Unfilled volume above bread: Open can and remove lid. Fill area between bread and side of can with rape seed, stopping when seed is level with top of bread (do not measure this volume). Fill a graduate with a measured quantity of rape seed and pour into remaining space above bread until can is level full. Level top of can with a straight edge, returning to graduate all seed falling over edge of can. determine volume by difference.

Can volume: Remove seed and bread from can and measure total can volume with rape seed.

Determine percent of unfilled volume above bread by the following formula:

Percent of unfilled volume above bread =

$$\frac{\text{Unfilled volume above bread minus Volume above unopened can lid}}{\text{Can volume minus Volume above unopened can lid}} \times 100$$

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:38Z
Check the source to verify that this is the current version before use.

4.5.5 Product testing. Chemical analysis for pH, moisture, and water activity (Aw) in the product shall be made in accordance with the following methods from the Official Methods of Analysis of the Association of Official Analytical Chemists. A total of 12 randomly selected samples per lot shall be tested for pH, moisture and Aw requirements. Any test result failing to conform to the applicable requirements in 3.4.1 and 3.4.2 shall be cause for rejection of the lot.

Test Finished product	Source	Method
pH	Chapter: Cereal Foods Section: Wheat Flour	Hydrogen-Ion Concentration Electrometric method.
Moisture 1/	Chapter: Cereal Foods Section: Bread	Total Solids in Entire Loaf of Bread (method applicable to bread not containing fruit).
Aw	Chapter: Vegetable Prods. Section: 32.004	Total Aw of Crust & Crumb

1/ Percent moisture equals 100 minus percent total solids.

4.5.6 Packaging examination. The examination shall be in accordance with the quality assurance provisions of MIL-C-43991.

5. PACKAGING

5.1 Preservation and packing. The product shall be preserved and packed in accordance with the applicable requirements of MIL-C-43991 (see 6.1).

5.2 Labeling and marking. Labeling of cans and marking of shipping containers shall be in accordance with MIL-C-43991.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article sample is required (see 3.1, 4.4, and 6.2).
c. Can size required (see 3.4.3).
d. Selection of the applicable level of preservation and packing (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 202610-16T12:38Z
Check the source to verify that this is the currentversion before use.

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection inspection, and approval of the first article.

6.3 Subject term (key word) listing.

Bread
Canned
Enriched
Rations, operational

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Project No. 8920-0469

U.S. GOVERNMENT PRINTING OFFICE: 1986 - 605-036/44379

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

U.S. ARMY NATICK RESEARCH, DEVELOPMENT
and ENGINEERING CENTER
ATTN: STRNC-ES
Natick, MA 01760-5014

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
U.S. Army Natick Research, Development
and Engineering Center
ATTN: STRNC-ES
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-1070E	Bread, White, Enriched, Canned

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP-1, Feb 86

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:38Z
Check the source to verify that this is the current version before use.